3D simulations of classical flocking models: a quantitative comparison with STARFLAG empirical data

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March 2026

Abstract

In this dissertation, we inquired into the behaviour of three classical flocking models, namely Reynolds', Couzin's and Vicsek's, simulating the flight dynamics in a 3-dimensional infinite space. We analysed the trends of the correlation function relative to velocities' orientations and the flock's polarization over time in four different scenarios: when free, when attacked by a predator, when reacting to a wall and when reacting to a column. Almost all methods performed well regarding polarization, as expected. Qualitatively speaking, Reynolds' and Couzin's methods performed well, while Vicsek's did not. To obtain satisfactory results for all three, much time was spent on manual tuning of the parameters. This is, in itself, a great limit of the models. Reynolds' model exhibited erratic behaviour: the dependency on the initial conditions determined non-replicable results in some scenarios, while it accomplished a somewhat close behaviour to the expected two-domain one in the others. The correlation function of a flock simulated using Vicsek's model was more satisfactory in the scenarios which were also the worst in regard to their qualitative performance. Couzin's model was the one that obtained the best results. Its correlation functions exhibited the correct behaviour in every scenario but the predator one. Still, Couzin's model is, akin to Reynolds', sensitive to the initial conditions. The main takeaway from the analysis is that neither polarization nor the correlation function works as a way to identify a flock's dynamics, since some flocks exhibited fine qualitative behaviour, while also performing poorly when tested against the empirical data, and vice versa; therefore, those metrics are not enough to properly assess the correctness of a flocking model's behaviour.

1 Introduction

Tra le rossastre nubi
Stormi d'uccelli neri,
Com'esuli pensieri,
Nel vespero migrar.

Among the reddish clouds
flocks of black birds,
like exhiled thoughts,
migrate in the verspers.

– G. Carducci (Literal translation by Tom Sharp)

The behaviour of flocking birds has been a subject of fascination for the human race for as long as anyone can remember. The study of such behaviour in a scientific way is much more novel; a flock of birds is treated as active matter [18], and as such uses the techniques of the field, which is a novel one, since its progress has been limited by computational capability and other material constraints. Speaking of material constraints, for much of the history of the field there was no available data to study the phenomenon of collective motion of birds. The first model to simulate it, called "boids", was created in 1987 by Craig Reynolds, a computer graphics researcher specialized in movies and videogames, and was based on nothing but reasonable assumptions and qualitative evaluation [15] [16].

In the following years, the field evolved towards a more scientific approach. Jump-started by T. Vicsek with his work on active matter (that has since been disputed since it operated in a toroidal space [11]), which was a simplified "boids" model created for the simulation of self-propelled particles [9] [12] [18], the study of collective motion saw a rapid development with the creation of many models; first, Y. Tu and J. Toner applied fluid dynamics to the system, obtaining a rigorous theoretical model based on the Navier-Stokes equation [13]. Then, I.D. Couzin developed the so-called "zonal" model, which was a refined Reynolds' model [7]. But alas, as scientific as those works might have been, they were still based on nothing but reasonable assumptions and qualitative evaluation, or, when it was quantitative, it was still testing reasonable quantities, not empirically verified ones.

That abruptly changed when, in 2008, A. Cavagna and his group published the first works about the STARFLAG system [2] [3] which set out, successfully,

to build a comprehensive dataset on collective animal behaviour; in particular, they recorded a great number of flocks of the common starling (*Sturnus vulgaris*) and their velocity, finally building the dataset the field needed. This group in particular was at the forefront of the field, first in analyzing their own dataset and finding the first emergent quantity of flocks, namely the shape of the so-called correlation function (of which the study determined that the interactions are scale-free) [4], then by analyzing it using statistical mechanics and by consequently using the MaxEnt statistical model to prove how the large scale interactions in a flock are determined by the small scale ones [10], and finally by building a predictive dynamical model akin to a Hamiltonian formulation for rotations [5]. In the meantime, their dataset sparked interest from other groups, thus leading to other dynamical models such as H. Hildenbrandt's one, which is a refined "zonal" model [6], in turn influenced by M. Bellerini's finding that birds do not follow metric rules but topological ones, as they react only to the 7 or 6 nearest flockmates [8].

In this work, we test the pre-2010 models in light of the findings by the Rome group, in particular that the shape of the correlation function is an emergent property of flocks [4]. To be more precise, we implement Reynolds', Vicsek's and Couzin's models, with predator and obstacle avoidance, and we test the similarity between their correlation function and the empirical one. The rationale for choosing the aforementioned models is that the post-2010 ones have already been tested in their respective seminal papers [5] [6], and the Toner-Tu model has technically already been tested by A. Cavagna, since his 2014 model reduces to Toner-Tu's formulation for large enough flocks [5]. Finally, Chaté's model [11], which is the one created when disputing Vicsek's one, was tested by Cavagna's group [10].

2 Literature overview

In this section, we cover the fundamental literature relative to the problem of the modeling of a flock of birds. We will follow an historical point of view, starting from C. Reynolds' pioneering "boids" model to Vicsek's active matter one and Couzin's "zonal" model. We then cover the importance of Cavagna et al.'s findings regarding the correlation function. In order for the discourse to flow conceptually, we must first cover the topic of emergence, perfectly crystallized by the famous essay "More is different" by P. W. Anderson [1].

2.1 Many-body physics and emergence

The reductionist hypothesis has, in the history of science, reigned sovereign as the basis of scientific inquiry; the possibility to reduce (hence the name) everything to a result of a number of fundamental laws is, self-evidently, intrinsically captivating to the human spirit. Many-body physics is, to the reductionist worldview, a challenge in and of itself. The concept that governs this field is that of emergence, which is the manifestation of different, unpredictable symmetries from the ones that govern the fundamental parts of a many-body system. A classic example is that of crystals: built according to perfectly homogeneous laws, the crystal unpredictably displays new symmetries. The general rule is, however, even in this case, that the system is less symmetrical than the underlying structure would suggest; as ordered as it may be, a crystal is not perfectly homogeneous. More is different, in the words of P. W. Anderson, a concept that is perfectly explained in the closing of the eponymous article:

> *[...]I offer two examples from economics [...]. Marx said that quantitative differences become qualitative ones, but a dialogue in Paris in the 1920's sums it up even more clearly:*
>
> > *Fitzgerald: The rich are different from us.*
> > *Hemingway: Yes, they have more money.* [1]

2.2 The flock of birds

In light of the concepts covered in the last subsection, it should come as no surprise how large groups of animals have been studied as an example of emergence, considering how they behave.

The emergent property of a flock of birds that is first obvious to the eye is its capacity to move as one, and to react to predators and obstacles as such without dividing the flock. Since it is physically impossible for each and every bird to perceive each and every one of their flockmates, they must follow a set of simpler rules; the modeling of a flock of birds is inquiring into those rules. For most of the field's history, that meant reasonably guessing them and to evaluate, with a mixture of qualitative and quantitative methods, if the model behaved as a flock. This evaluation was intrinsically flawed, since it did not have real hard data to be tested against. This is where Cavagna's paper comes in handy. We will now explain three classical models for simulating a flock of birds, and then explain Cavagna's findings and how they can be used to test the aforementioned models.

2.2.1 Reynolds' model: the "boids"

The first model, historically, is the one named "boids", by Craig Reynolds. He set out to animate bird-oid objects (hence the name "boids", although the reasons for the name are more than one, as stated in Appendix A) which moved based on a simple set of three rules:

- **Cohesion:** the boids tend to be attracted to the centroid of nearby flockmates;

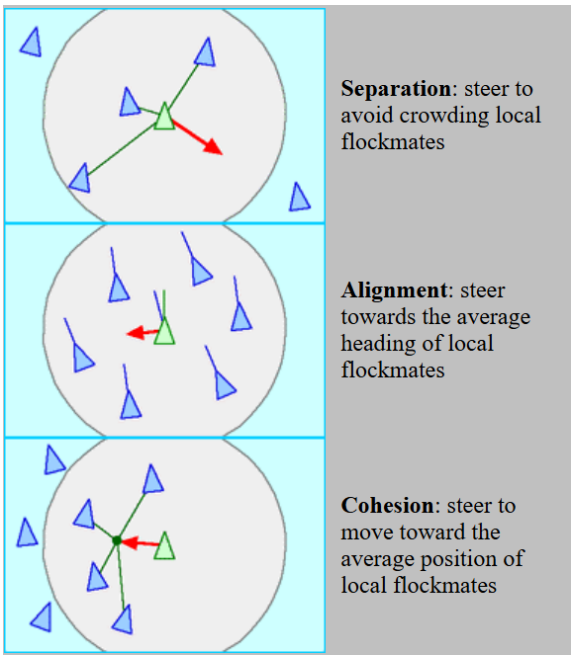


Figure 1: Pictorial representation of the boids' rules from Reynolds' personal site. [14]

- **Alignment:** the boids tend to align their direction with that of nearby flockmates;
- **Separation:** the boids tend to be repelled by nearby flockmates;

It should be noted that the distance that defines "nearby" is the same for all three of the forces, as suggested by Figure 1.

Reynolds also used some more rules in order for the model to behave more realistically [16], some of which were later refined in a more recent paper [17]:

- **Maximum speed:** in order to ensure that no boid gets too fast too quickly and escapes the flock, the speed gets capped;
- **Minimum speed:** in order to ensure that the flock does not become stagnant, no boid can get slower than an arbitrary threshold;
- **Limited acceleration:** to simulate more realistic behaviour, boids can accelerate only for a limited amount;
- **Limited field of view (FOV):** to simulate more realistic behaviour, every boid has a limited peripheral perception, namely it does not react to the boids behind it;

The limited steering actually has another effect, considering the way Reynolds explains the implementation of the model; namely, he implements each rule's effect on an accumulator, and gives an order to them, specifically separation, alignment and cohesion. What this means, operatively, is that if the sum of two or three forces exceeds the magnitude given by the limited steering, only the last one or two applied will be rescaled. Also, if either alignment or separation exceeds the maximum steering and so saturate the accumulator, cohesion does not operate on the boid. In the latter case, alignment does not operate either.

Considering that he is an Academy award winning computer graphic scientist, specialized in the simulation of natural phenomena for the entertainment industry [15], his model only aims at a qualitative representation of a flock: that means that the only test he truly cares about is the eye test, as stated in the original '87 paper [16]. This is a key detail for our dissertation.

2.2.2 Vicsek's model: self-propelled particles

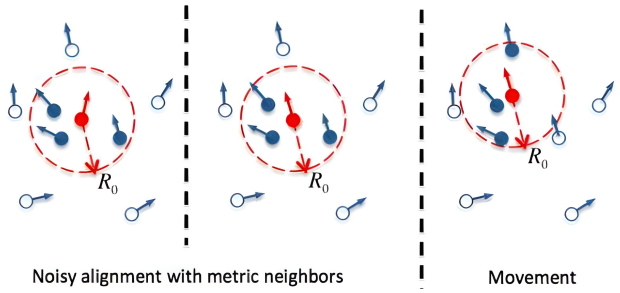


Figure 2: Representation of the behaviour of a particle in Vicsek's model. [12]

A simplification of Reynolds' model, Vicsek's was created as a mean of studying active matter, as the name of the original paper suggests. Vicsek's interest lies in phase transitions in systems of self-propelled particles. Consequently, his model tried to emulate this kind of system with a toroidal space. [12]

Derivatively from the alignment constraint in the "boids" model, Vicsek's is based on one simple rule; from the words of the man himself:

> *The only rule of the model is: at each time step a given particle driven with a constant absolute velocity assumes the average direction of motion of the particles in its neighbourhood of radius r with some random perturbation added*, as shown in Figure 2. [9]

The aforementioned rule, when calling the random noise $\Delta\theta$, reads as:

$$\begin{aligned} \theta(t+1) &= \langle\theta(t)\rangle_r + \Delta\theta \\ \langle\theta(t)\rangle_r &= \frac{1}{N_{neighbours}} \sum_{|x_j - x_i| \leq r} \theta_j(t) \end{aligned} \tag{1}$$

His model does have one additional constraint, namely uniform speed [12]; it should also be noted that since technically it is not trying to emulate birds but particles, all of the rules that apply to other models to simulate realistic behaviour (i. e. limited FOV, limited steering) do not make sense for Vicsek's model.

By simulating a system that obeyed Equation 1, Vicsek was able to observe phase transitions and flocking; in other words, the system exhibited order from a not ordered starting point [9]; as much as the model is not based on the behaviour of flocks of birds, it does nonetheless exhibit similar collective patterns as them, as Vicsek claims (although that was later attributed to the use of a toroidal space [11]), which is why we decided to include it in this work.

Since Vicsek is a physicist, he does perform quantitative analysis, but considering that the scope of his work is different than ours, they are not relevant for our own dissertation.

2.2.3 Couzin's model: the "zonal" algorithm

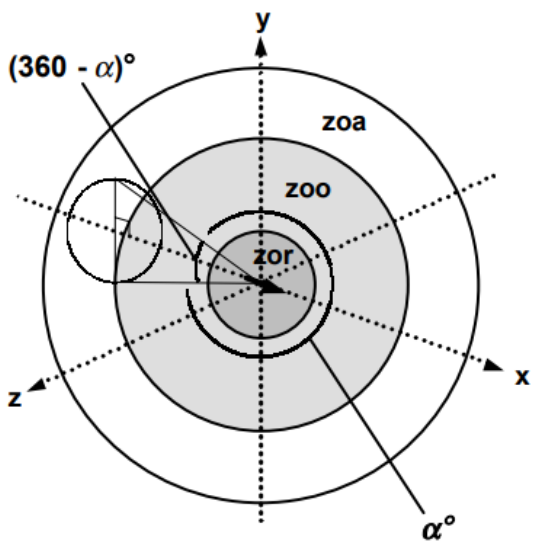


Figure 3: Representation of an individual in the model centered in the origin. *zoo* stands for zone of orientation, *zoa* for zone of attraction and *zor* for zone of repulsion.

An advancement more from Reynolds' than from Vicsek's model is the one by Ian D. Couzin, often referred to as the "zonal" model for reasons that will be self-evident in a moment.

Couzin, like Reynolds, asserts that every bird should be dominated by three forces: attraction, orientation and repulsion, which respectively correspond to cohesion, alignment and separation defined for the "boids" model in subsubsection 2.2.1. The similarity to Vicsek's model, instead, lies in the fact that the speed is fixed.

Couzin defines the forces' effects algebraically, more specifically every interaction should determine the direction in the following time step as follows:

$$\mathbf{d}_r(t+\tau) = -\sum_{j\neq i}^{n_r} \frac{\mathbf{r}_{ij}(t)}{|\mathbf{r}_{ij}(t)|} \tag{2}$$

$$\mathbf{d}_o(t+\tau) = \sum_{j=1}^{n_o} \frac{\mathbf{v}_j(t)}{|\mathbf{v}_j(t)|} \tag{3}$$

$$\mathbf{d}_a(t+\tau) = \sum_{j\neq i}^{n_a} \frac{\mathbf{r}_{ij}(t)}{|\mathbf{r}_{ij}(t)|} \tag{4}$$

The key difference from Reynolds' model lies in when the forces act. Couzin defines three zones, one for every interaction. The zone of repulsion is the one with the smaller radius, and as such is a spherical sector, in order to account for the bird's limited FOV. The zones of orientation and attraction are instead sectors of spherical shells (again to emulate the limited FOV), and the zone of attraction is the largest one, as shown in Figure 3. When a bird has any neighbours in the zone of repulsion, Equation 2 applies for the whole set of n_r neighbours. Repulsion takes precedence: if and only if $n_r = 0$ the other forces may apply. In particular, if there is a number of neighbours only in one of the other zones, only the corresponding force applies, if there are some in both, $\mathbf{d}(t+\tau) = \frac{\mathbf{d}_o(t+\tau)+\mathbf{d}_a(t+\tau)}{2}$ [7].

Couzin, akin to Vicsek, is a scientist (a biologist to be precise) and as such he performs some qualitative analysis of his model. We will not cover them extensively because the analysis by Cavagna's group is more thorough.

2.3 Cavagna's findings: the first experimental dataset and the first emergent property

In 2008, it was found that the flocking birds' criteria for interaction are not metric, but topological: specifically, birds interact with their closest six or seven neighbours [8]. Cavagna and his group, by using a dataset of which the specifics can be read at [2] [3], wanted to refine that claim. Specifically, they wanted to understand the scale of the system's correlation. Cavagna asserts that the main emergent behaviour in a flock is not its order, but its collective reaction, be it to predators or obstacles; his hypothesis is that it cannot possibly be a result of a movement that is correlated with just the six or seven closest neighbours' one [4]. A didactic example, in this sense, is that of the "telephone" game, in which one says something and their neighbour communicates said thing to their own neighbour, and so on; if each player communicates effectively, although each one only directly communicates with their neighbours, the final message is correlated with the initial one. The key takeway from the telephone game example is that interaction range and correlation range could be different, even dramatically so.

Cavagna, differently from his predecessors, has empirical data: that means that his analysis is rooted in (empirical) reality, and he can consequently prove (or disprove) his hypothesis. By analyzing the data, his group observed how the velocity fluctuations tended

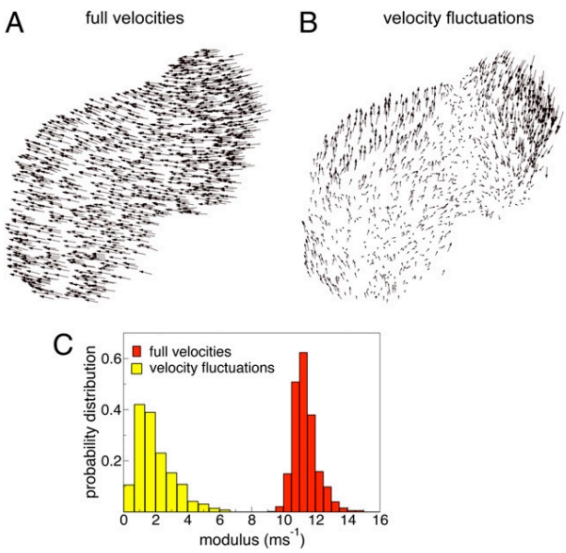


Figure 4: 2d projections of an arbitrary flock's velocity and velocity fluctuations, combined with their distribution.

to be grouped in two regions of the flock, as seen in Figure 4. In order to crystallize this observation mathematically, his group defined the correlation function $C(r)$ as:

$$C(r) = \frac{1}{c_0} \frac{\sum_{ij} \vec{u}_i \cdot \vec{u}_j \, \delta(r - r_{ij})}{\sum_{ij} \delta(r - r_{ij})} \tag{5}$$

where $\vec{u}_i = \vec{v}_i - \frac{1}{N}\sum_{k=1}^{N} \vec{v}_k$, is the fluctuation around the mean flock's velocity for each bird i, $\delta(r - r_{ij})$ is a smoothed Dirac δ-function selecting pairs of birds at mutual distance r; $\vec{u}_i \cdot \vec{u}_j = u_i^x u_j^x + u_i^y u_j^y + u_i^z u_j^z$ and c_0 is a normalization factor (of dimension $\mathrm{m}^2 \cdot \mathrm{s}^{-2}$) such that $C(r = 0) = 1$ [4].

By looking at the same image, one can see how the velocities, differently from their fluctuations, are well aligned. To quantify this global ordering, they inquired about the flock's polarization, which is defined as such:

$$\Phi = \left|\left| \frac{1}{N} \sum_{i=1}^{N} \frac{\vec{v}_i}{||\vec{v}_i||} \right|\right| \tag{6}$$

In the paper, they obtain an average value of $\Phi = 0.96 \pm 0.03$ [4].

The correlation function, as defined in Equation 5, should span between 1 and -1, and positive values represent positive correlation while negative ones represent anti-correlation. A single-signed section should represent a subgroup of the flock that reacts in the same way.

By analyzing Figure 5, we can finally infer what Cavagna's paper asserts with greater theoretical rigour: the flock's correlations are scale-free, and all flocks behave the same, as ξ, the correlation length, grows linearly with their size. The non-trivial part is that the

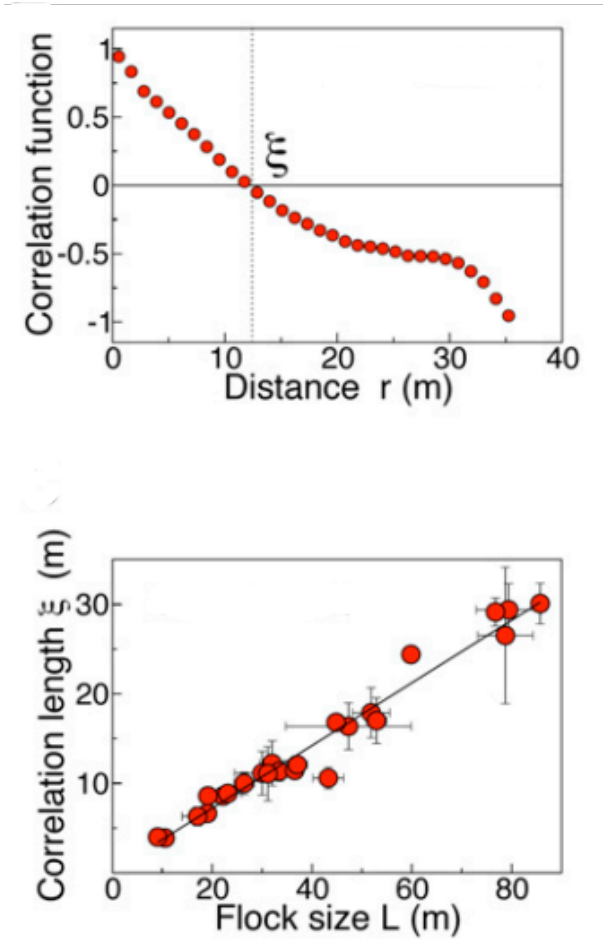


Figure 5: Plotting of the correlation function of a flock, and of the intercept ξ of many flocks' correlation function against their size.

correlation function $C(r)$ only has two single-signed sections, which can heuristically explain how the interactions are scale-free: the whole flock reacts to each and every bird's movement either in a positive way or a negative one; in other words, there are only two domains of reaction: either compliant or opposite [4].

The claim is refined, in a later paper, by employing the methods of statistical mechanics. With the idea in mind to find a statistical model for the flock, they used the maximum entropy (MaxEnt) principle, which states that the most assumption-less model (not necessarily the most correct one) is the one that maximises the entropy of the system, given some constraints. In this case, the MaxEnt distribution is:

$$P(\{\vec{s}_i\}) = \frac{1}{Z(\{J_{ij}\})} \exp\left[\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\vec{s}_i \cdot \vec{s}_j\right] \tag{7}$$

Where $Z(\{J_{ij}\})$ is the proper normalization factor, and J_{ij} is calculated in order for the predicted correlation function to be the same as the empirical one; this is one of our constraints. The most logical simplification of Equation 7 is the one in which every boid interacts

with a limited number of neighbours n_c with the same intensity (or weight) J:

$$P(\{\vec{s}_i\}) = \frac{1}{Z(J, n_c)} \exp\left[\frac{J}{2}\sum_{i=1}^{N}\sum_{j\in n_c^i} \vec{s}_i \cdot \vec{s}_j\right] \tag{8}$$

where $j \in n_c^i$ means that bird j belongs to the first n_c nearest neighbors of i. As it turns out, this model is incredibly predictive, and confirms once again that the interactions are scale-free and most importantly topological, as proved by the fact that the n_c that maximizes the $P(\{\vec{s}_i\})$ in Equation 8 is steady in regards to the size of the analyzed flock. [10]

The gist of the whole analysis by Cavagna's group is that the interactions between birds in a flock are indeed topological and not metric, but while finding that out, they discovered the emergent properties that we will analyze for other, less recent models.

3 Code implementation

In order to run the simulations for the three models and compute all the statistics presented above, a custom Python project was developed. It is structured into six modules:

- `config.py`: consists of data classes storing the constant parameters of the simulation;
- `simulation.py`: contains the core logic of the simulation;
- `animation.py`: fully dedicated to the graphic representation of the flock dynamics;
- `statistics.py`: handles the computation of correlation and polarization functions;
- `auxiliary_functions.py`: contains miscellaneous functions, mainly related to plotting;
- `main.py`: manages the options and the overall project workflow.

This section focuses exclusively on the `simulation.py` module, as it is the most relevant to the purpose of this report. For the sake of clarity, rather than performing a line-by-line code review, we will outline the fundamental logic of this module. For the full code, please refer to Appendix B.

At the beginning of the file, two classes are defined in order to handle the initialization phase:

- `FlockState`: initializes the boids' positions and velocities;
- `PredatorState`: initializes the predator's position and velocity.

Following the initialization structures, the core computation function is presented:

```
def compute_distances_and_fov(pos, vel):
    dist_vects = pos[:, np.newaxis, :] -
    ↪  pos[np.newaxis, :, :]
    dist_norms = np.linalg.norm(dist_vects,
    ↪  axis=-1)
    dot_prods = (vel[:, np.newaxis, :] *
    ↪  (-dist_vects)).sum(axis=-1)
    vel_norms = np.linalg.norm(vel,
    ↪  axis=-1)[:, np.newaxis]
    cos_angles = dot_prods /
    ↪  np.maximum(vel_norms * dist_norms,
    ↪  1e-9)
    return dist_vects, dist_norms, cos_angles
```

This function computes the distance vectors, the distance norms, and the viewing angles between each pair of boids in the flock. Instead of employing a classical `for` loop to iterate over the elements of the flock's position matrix, the computation is highly optimized through `NumPy` array broadcasting. The objects returned by this function are extensively used throughout the code.

Subsequently, three distinct functions are defined to simulate the tested models: `compute_reynolds`, `compute_couzin`, and `compute_vicsek`. The architecture of these functions is structurally similar. First, a boolean mask matrix is defined to identify each boid's neighbors within its specific action range and field of view:

```
def compute_reynolds(pos, vel, dist_vects,
↪  dist_norms, cos_angles, predator_state):
    mask = (dist_norms <
    ↪  cfg.reynolds_const.action_range) &
    (dist_norms > 0) & (cos_angles >
    ↪  cfg.glob_const.cos_fov)
    mask_3d = mask[:, :, np.newaxis]
    n_neighbors = mask.sum(axis=1)
    has_neighbors = n_neighbors > 0
    ...
```

Then, the forces governing each model's interactions are computed according to their original mathematical formulations:

```
def compute_reynolds(pos, vel, dist_vects,
↪  dist_norms, cos_angles, predator_state):
    ...
    # Cohesion
    sum_pos = (pos[np.newaxis, :, :] *
    ↪  mask_3d).sum(axis=1)
    centroid = sum_pos[has_neighbors] /
    ↪  n_neighbors[has_neighbors,
    ↪  np.newaxis]
    neighb_dist = centroid -
    ↪  pos[has_neighbors]
    neighb_dist_norm =
    ↪  np.linalg.norm((neighb_dist),
    ↪  keepdims=True, axis=1)
```

```
    coh_vel[has_neighbors] = (neighb_dist) *
    ↪  \
        cfg.reynolds_const.coh_par /
        ↪  (neighb_dist_norm**3)
    # Alignment
    sum_vel = (vel[np.newaxis, :, :] *
    ↪  mask_3d).sum(axis=1)
    mean_vel = sum_vel[has_neighbors] /
    ↪  n_neighbors[has_neighbors,
    ↪  np.newaxis]
    ali_vel[has_neighbors] = (
        mean_vel - vel[has_neighbors]) *
        ↪  cfg.reynolds_const.ali_par
    # Separation
    safe_distance_sq = np.where(dist_norms ==
    ↪  0, 1.0, dist_norms**3)
    repulsion = dist_vects /
    ↪  safe_distance_sq[:, :, np.newaxis]
    sep_vel = (repulsion *
    ↪  mask_3d).sum(axis=1) *
    ↪  cfg.reynolds_const.sep_par
    ...
```

Interactions related to obstacle and predator avoidance are handled globally by two separate functions, which can then be integrated into any of the three models:

```
def compute_obstacle_avoidance(flock_pos):
    obs_dist_vects = flock_pos[:, np.newaxis,
    ↪  :] -  cfg.obstacles_const.positions
    [np.newaxis, :, :]
    obs_dist_norms =
    ↪  np.linalg.norm(obs_dist_vects,
    ↪  axis=-1)
    mask = (obs_dist_norms <
    ↪  cfg.obstacles_const.action_range) & (
        obs_dist_norms > 0)
    mask_3d = mask[:, :, np.newaxis]
    safe_distance_sq =
    ↪  np.where(obs_dist_norms == 0, 1.0,
    ↪  obs_dist_norms**2)
    repulsion = obs_dist_vects /
    ↪  safe_distance_sq[:, :, np.newaxis]
    obs_avoid_vel = (repulsion *
    ↪  mask_3d).sum(axis=1) * \
        cfg.obstacles_const.rep_par
    return obs_avoid_vel
```

```
def compute_predator_avoidance(flock_pos,
↪  pred_pos):
    pred_dist_vects = flock_pos[:,
    ↪  np.newaxis, :] - pred_pos[np.newaxis,
    ↪  :, :]
    pred_dist_norms =
    ↪  np.linalg.norm(pred_dist_vects,
    ↪  axis=-1)
    mask = (pred_dist_norms <
    ↪  cfg.predator_const.dist_par) & (
        pred_dist_norms > 0)
    mask_3d = mask[:, :, np.newaxis]
    safe_distance_sq =
    ↪  np.where(pred_dist_norms == 0, 1.0,
    ↪  pred_dist_norms**2)
    repulsion = pred_dist_vects /
    ↪  safe_distance_sq[:, :, np.newaxis]
    pred_avoid_vel = (repulsion *
    ↪  mask_3d).sum(axis=1) * \
        cfg.predator_const.sep_par
    return pred_avoid_vel
```

Finally, each model processes and applies these spatial forces differently:

- in `compute_reynolds`, each acceleration request is accumulated and constrained within a finite capacity using a clamping function;
- in `compute_couzin`, the zonal priority model is applied, and a spherically wrapped Gaussian noise is added to the target direction;
- in `compute_vicsek`, the alignment force is applied to each boid, followed by the addition of uniform angular noise.

The final section of the `simulation.py` module is devoted to the `update_flock` function. This function acts as the main controller, responsible for applying the selected method to compute the velocity updates between two subsequent time steps:

```
def update_flock(flock_state: FlockState,
↪  predator_state: PredatorState, method:
↪  str):
    dist_vects, dist_norms, cos_angles =
    ↪  compute_distances_and_fov(
        flock_state.pos, flock_state.vel)
    match method.lower():
        case "reynolds":
            boids_prov_vel =
            ↪  compute_reynolds(
                flock_state.pos,
                ↪  flock_state.vel,
                ↪  dist_vects, dist_norms,
                ↪  cos_angles,
                ↪  predator_state)
        case "couzin":
            boids_prov_vel = compute_couzin(
                flock_state.pos,
                ↪  flock_state.vel,
                ↪  dist_vects, dist_norms,
                ↪  cos_angles,
                ↪  predator_state)
        case "vicsek":
            boids_prov_vel = compute_vicsek(
                flock_state.pos,
                ↪  flock_state.vel,
                ↪  dist_norms,
                ↪  predator_state)
```

```
        case _:
            raise ValueError(
                f"Method '{method}' is
                ↪ invalid. Choose among
                ↪ 'reynolds', 'couzin' or
                ↪ 'vicsek'.")
    flock_state.vel += boids_prov_vel
    ...
```

After the specific model's logic is computed, the boids' velocity vectors are updated and, if required by the model (e.g., Reynolds), clamped to enforce physical speed limits. Finally, an `if` statement is employed to manage the predator dynamics, updating its position based on the flock's centroid attraction while enforcing its own maximum speed constraints:

```
    ...
    if cfg.commands.predator_bool == True:
        pred_prov_vel = predator_move(
            flock_state.pos,
            ↪ predator_state.pos)
        predator_state.vel += pred_prov_vel
        # Predator's velocity limit
        pred_speed =
        ↪ np.linalg.norm(predator_state.vel,
        ↪ axis=1, keepdims=True)
        predator_state.vel = np.where(
            pred_speed >
            ↪ cfg.predator_const.max_speed,
            (predator_state.vel / pred_speed)
            ↪ *
            ↪ cfg.predator_const.max_speed,
            predator_state.vel
        )
        predator_state.pos +=
        ↪ predator_state.vel

    flock_state.pos += flock_state.vel
```

4 Data collection and analysis

In this section, we cover how we collected the data using the aforementioned program and we analyse the numerical results. For more data, as well as some graphical renders of the moving flock, please refer to Appendix B.

4.1 Methodology

Remembering the nature of the methods, which is qualitative, we established the following protocol for data collection:

1. We fixed the following parameters:
   - **number of boids** = 200, 500 and 1000;
   - **field of view** = 290°;
   - **range for the boids' initial positions** = $[-\frac{\text{number of boids}}{40}, \frac{\text{number of boids}}{40}]$;
   - **number of timesteps analysed** = 500;
   - **range of speed for the predator** = [3.0, 25.0];
   - **maximum change in speed for the predator** = 1.5;
   - **attraction parameter for the predator** = 2.0;
2. We performed, for every method and for the three numbers of boids listed, manual tuning of the free parameters by testing their qualitative output, which was in turn checked by producing a video or a gif of the duration of 500 timesteps;
3. After finding the adequate parameters, we collected the correlation function and the polarization over time for the flock when free, attacked by a predator and approaching an obstacle, be it a wall or a column.

4.2 Reynolds' method: results and comments

4.2.1 Results

The results below were obtained by employing the parameters listed in Table 1.

Parameter	200 boids	500 boids	1000 boids
Cohesion	5.5	5.5	5.5
Alignment	0.050	0.20	0.80
Separation	0.025	0.10	0.50
Action range	20	20	20
Speed range	1.0-4.0	1.0-4.0	1.0-4.0
Maximum delta	0.25	0.25	0.25

Table 1: Parameters found by manual tuning for Reynolds' model.

Reynolds' model proved itself to be, as far as quantitative analysis goes, not predictable and rather sensitive to the initial conditions (which are the only random variable of the method). As expected, polarization stayed stably close to one, with the necessary oscillations deriving from the situations (for example, when hitting a wall it briefly decreased); such results can be inspected regarding the 1000 boids flock in Figure 6, but are consistent regardless of the number of boids.

The same cannot be said regarding the correlation function. Although in our analysis we did find some that exhibited the already explained two domain behaviour, they were not the norm; in general, the typical correlation function was more akin to a random oscillation around 0, as shown in Figure 7.

4.2.2 Comments

While manually tuning the model, we noticed how sensitive Reynolds' model is to changes in the free parameters, and how unsuited for quantitative analysis it is, since the correlation function tends to be unstable over different iterations of the model. It is also, as stated before, remarkably sensitive to the initial conditions. Qualitatively, on the other hand, it performed well.

4.3 Vicsek's method: results and comments

4.3.1 Results

The following results were obtained by employing the parameters listed in Table 2.

Parameter	**200 boids**	**500 boids**	**1000 boids**
Speed	2.0	2.0	2.0
Action range	10	20	30
Angular noise	1.0°	1.0°	1.0°

Table 2: Parameters found by manual tuning for Vicsek's model.

As was predictable, Vicsek's model, qualitatively, does not perform well. It obtains reasonably acceptable (albeit very artificial looking) simulations for free and wall-approaching flocks, but when attacked by a predator or approaching a column the flock does not behave correctly, and splits into two or more sub-flocks.

Quantitatively speaking, the polarization function is predictably close to one (the model concerns only velocity alignment, after all), although it should be noted that a decrease is clearly noticeable when the flocks split, as can be seen in Figure 8.

The correlation function exhibits the correct behaviour both in the case of a flock attacked by a predator and reacting to a column, as seen in Figure 9, but these results have to be paired with the bad qualitative output.

4.3.2 Comments

Vicsek's model, while less sensitive to parameters than Reynolds', has its own share of problems, both quantitative and qualitative. The most important ones are the latter, since Vicsek's model, as stated before, is not fit for this type of analysis. Quantitatively, its problems lie in the fact that, while polarization exhibits good results (apart from the case of a predator attacking the flock), the correlation function only exhibits the correct behaviour in two of the four analysed scenarios, which are also the ones in which the model qualitatively performs worse.

4.4 Couzin's method: results and comments

4.4.1 Results

The following results were obtained by employing the parameters listed in Table 3.

Parameter	**200 boids**	**500 boids**	**1000 boids**
Speed	2.0	2.0	2.0
ZOA	20	25	40*
ZOO	10	20	25
ZOR	1	1.5	3
Maximum steering angle (free)	1.0°	1.0°	1.0°
Maximum steering angle (predator)	5.0°	7.5°	5.0°
Maximum steering angle (column)	3.0°	1.0°	1.0°
Maximum steering angle (wall)	3.0°	3.5°	3.0°
Angular noise	0.1°	0.1°	0.1°

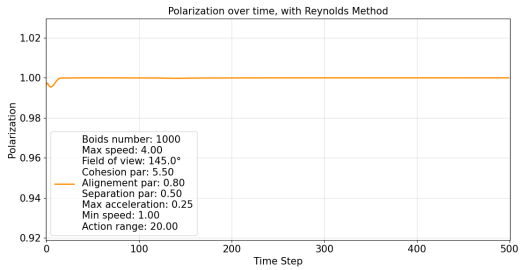
Table 3: Parameters found by manual tuning for Couzin's model.
*For the 1000 boids simulation regarding the reaction to a column, the ZOA was set to 35.

Couzin's model, although more sensitive to the parameters than the others (as the reader may infer by looking at the tables), is the one that both qualitatively and quantitatively performs best. As for the other methods, polarization is stably close to one, with the necessary oscillations deriving from the situations; such results can be inspected regarding the 1000 boids flock in Figure 10, but are consistent regardless of the number of boids.

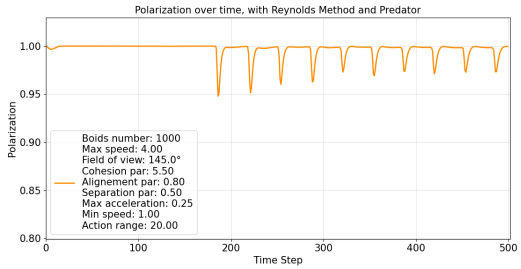
The correlation function, as shown in Figure 11, performs well in all of the analyzed scenarios but one, namely the one in which the flock is attacked by a predator; the reader should be made aware that for the 1000 boids flock the correlation function did actually exhibit the desired two-domain behaviour, but the pattern of increasing again after a certain distance was consistent with that of the other simulations' correlation functions.

4.4.2 Comments

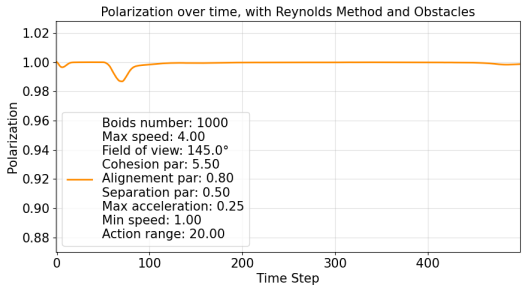
As expected, considering how Couzin's model is the most advanced one, it is the one that performs best, both quantitatively and qualitatively. That does come with a downside: the sensitivity to the parameters is comparable (a bit higher, to be exact) to that of Reynolds' model, and as such the model requires its fair share of manual tuning in order for it to work as desired; nonetheless, the results are still the best across the tested methods, even when taking into account the time required to manually tune the parameters.



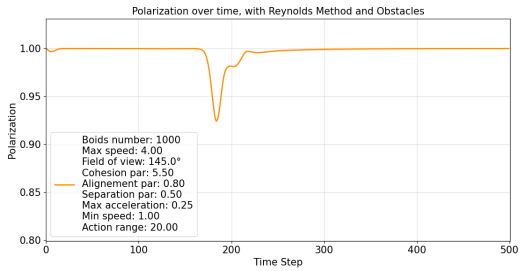
(a) Polarization over time for a free flock of 1000 boids.



(b) Polarization over time for a flock of 1000 boids while attacked by a predator.

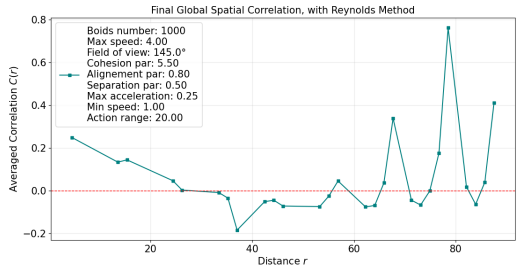


(c) Polarization over time for a flock of 1000 boids reacting to a column.

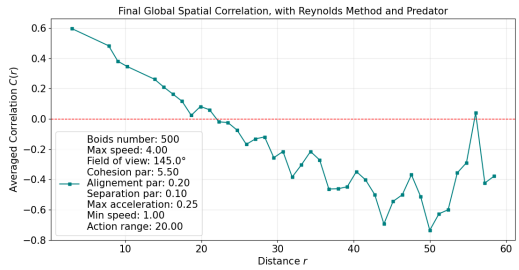


(d) Polarization over time for a flock of 1000 boids reacting to a wall.

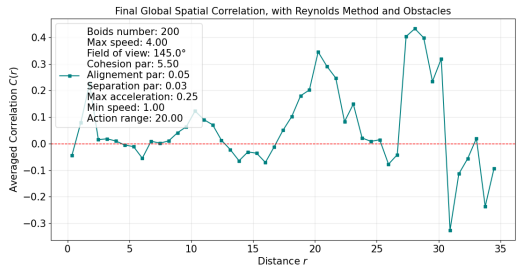
Figure 6: Polarization over time plotted in different situations for a flock simulated using Reynolds' model.



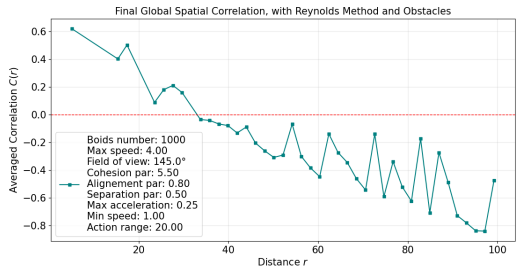
(a) Correlation function of a 1000 boids free flock.



(b) Correlation function of a 500 boids flock attacked by a predator.

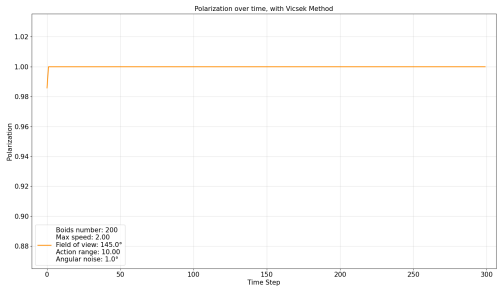


(c) Correlation function of a 200 boids flock reacting to a column.

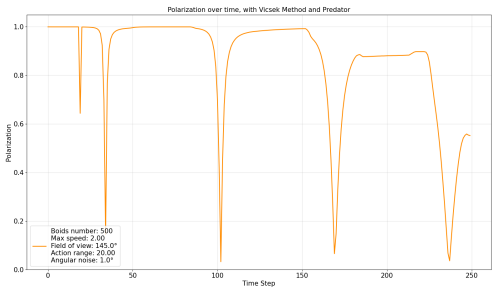


(d) Correlation function of a 1000 boids flock reacting to a wall.

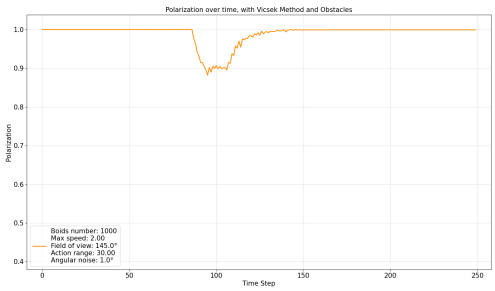
Figure 7: Correlation functions plotted in different situations for a flock simulated using Reynolds' model.



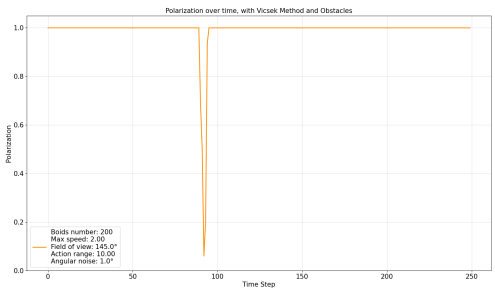
(a) Polarization over time of a 200 boids free flock.



(b) Polarization over time of a 500 boids flock while attacked by a predator.

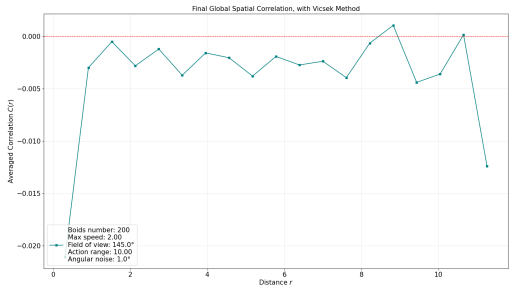


(c) Polarization over time of a 1000 boids flock reacting to a column.

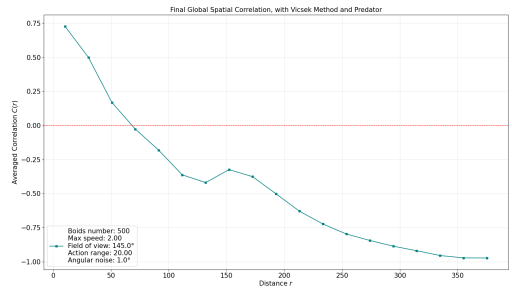


(d) Polarization over time of a 200 boids flock reacting to a wall.

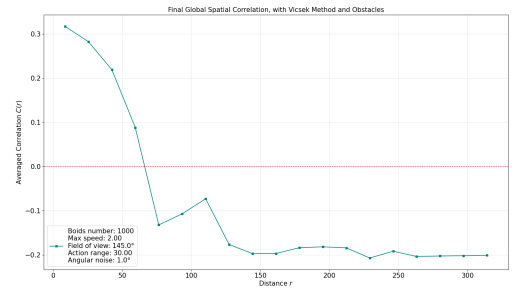
Figure 8: Polarization over time plotted in different situations for a flock simulated using Vicsek's model.



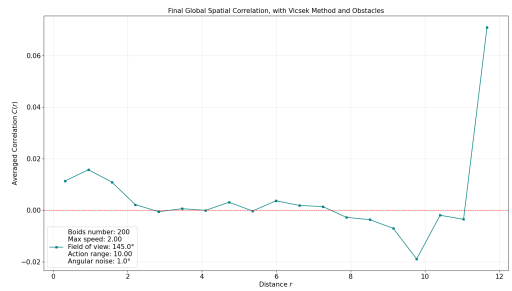
(a) Correlation function of a 200 boids free flock.



(b) Correlation function of a 500 boids flock while attacked by a predator.

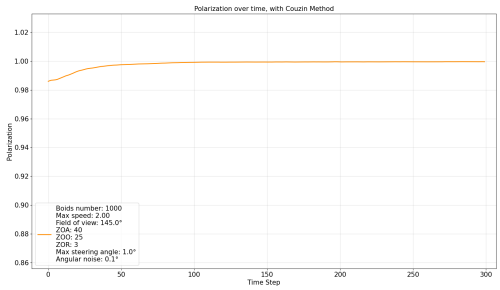


(c) Correlation function of a 1000 boids flock reacting to a column.

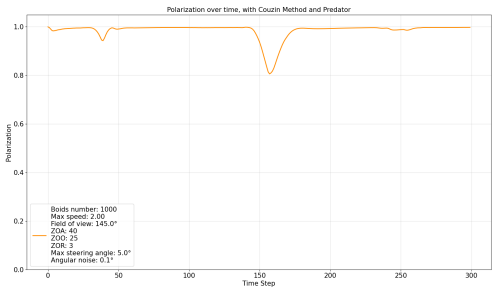


(d) Correlation function of a 200 boids flock reacting to a wall.

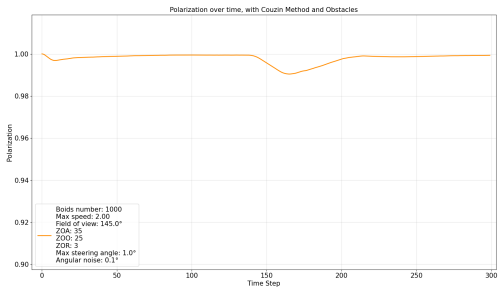
Figure 9: Correlation functions plotted in different situations for a flock simulated using Vicsek's model.



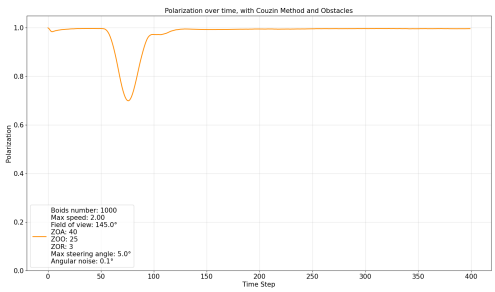
(a) Polarization over time for a free flock of 1000 boids.



(b) Polarization over time for a flock of 1000 boids while attacked by a predator.

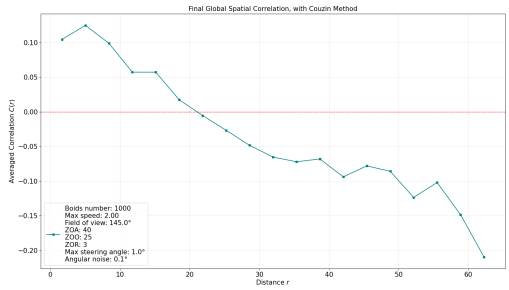


(c) Polarization over time for a flock of 1000 boids reacting to a column.

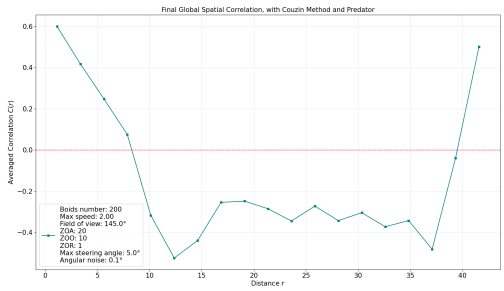


(d) Polarization over time for a flock of 1000 boids reacting to a wall.

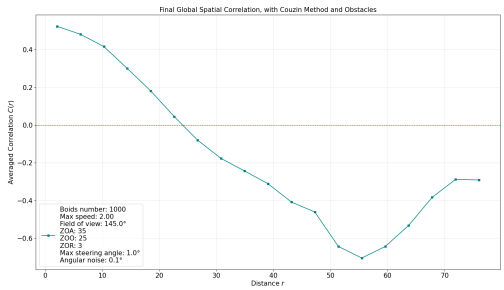
Figure 10: Polarization over time plotted in different situations for a flock simulated using Couzin's model.



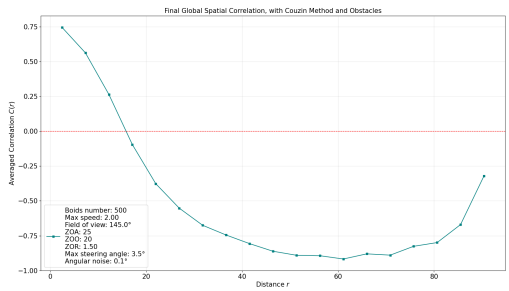
(a) Correlation function of a 1000 boids free flock.



(b) Correlation function of a 200 boids flock while attacked by a predator.



(c) Correlation function of a 1000 boids flock reacting to a column.



(d) Correlation function of a 500 boids flock reacting to a wall.

Figure 11: Correlation functions plotted in different situations for a flock simulated using Couzin's model.

5 Conclusions

In this dissertation, we analyzed how well classic flocking models performed against the empirical data obtained by Cavagna's group; more specifically, we inquired into the behaviour of their correlation function $C(r)$, as defined in Equation 5, and their polarization Φ, as defined in Equation 6, in four different scenarios: when free, when attacked by a predator, when reacting to a wall and when reacting to a column. In this section, we state our results and draw some conclusions regarding the aforementioned.

While analysing the data, we encountered some common results across methods and scenarios, and some specific ones. We will start by listing the former.

Firstly, all methods performed almost perfectly regarding polarization, with the exclusion of Vicsek's one when attacked by a predator; every method exhibited a polarization close to the empirical result of $\Phi = 0.96 \pm 0.03$ [4], each considering the necessary oscillations (or lack thereof) caused by the scenario. This is to be expected, given the importance that every method gave to the boids' alignment. Secondly, Reynolds' and Couzin's methods performed well, qualitatively speaking, while Vicsek's did not when reacting to a predator or a column. This result was also hardly surprising, since the latter method was conceived for a toroidal space, and as such does not concern itself with mutual attraction of the elements of the simulation, and the former two were conceived as qualitative methods, and as such their qualitative performance had to be good. To obtain satisfactory results for all three, though, much time was spent on manual tuning of the parameters. This is, in itself, a great limitation of the models, since their qualitative output is related to their complexity (and consequently the number of parameters), and manual tuning becomes more difficult once complexity rises.

We now move on to the more specific observations regarding each model.
Reynolds' model exhibited erratic behaviour in regard to the correlation function both when free and when reacting to a column. The dependency on the initial conditions determined non-replicable results in those scenarios, while it accomplished a behaviour similar to the expected two-domain one in the other two, with no dependency on the number of boids whatsoever. Qualitatively, it seemed to work best when the number of boids decreased. The latter result is explainable by the fact that, when Reynolds conceived the model, technology was not able to create many boids, and consequently he tested it on a limited number of them.

The correlation function of a flock simulated using Vicsek's model was more satisfactory in every scenario but the free flock and the wall one. Coincidentally, the methods with the best correlation functions were also the worst in regard to their qualitative output.

Couzin's model was the one that obtained the best results, as expected given its more advanced nature. Its correlation functions exhibited the correct behaviour in every scenario but the predator one; this, paired with the fact that it also obtained the best qualitative results, puts it as the best of the tested methods. The reader must, however, know that Couzin's model is, like Reynolds', sensitive to the initial conditions, and so may sometimes exhibit unexpected behaviour.

The main takeaway from the analysis is that neither polarization nor the correlation function works as a way to identify a flock's dynamics. Some flocks exhibited great qualitative behaviour, while also performing poorly when tested against the empirical data, and vice versa; as much as the aforementioned properties are emergent properties of a bird flock, they are not enough to properly assess the correctness of its behaviour.

6 Further developments

The three models implemented and tested in this work are faithful replicas of the original models' formulations, which implies that all of them are based on metric interactions. In these frameworks, each boid is influenced by all neighbors located within a physical radius. However, empirical observations by Bellerini et al. [8] have demonstrated that interactions within real flocks of starlings are topological rather than metric; specifically, each bird interacts with a fixed number of nearest neighbors, regardless of their physical distance. A highly relevant future development would entail modernizing these classical models by transitioning from metric to topological interaction rules. However, this paradigm shift comes at a significant computational cost: identifying topological neighbors requires continuously sorting and ranking the elements of the distance matrix at each time step, thereby increasing the algorithm's complexity.

Furthermore, the simulation framework developed herein provides a robust foundation for more advanced statistical analyses. Future work could expand the current metrics to investigate the presence of scale-free phenomena, primarily by verifying whether the correlation length scales linearly with the overall flock size. Additionally, the analysis could be enriched by computing a speed correlation function, as opposed to solely analyzing orientational (directional) correlation [4]. This was not done in this work since the tested models do not regularly exhibit the necessary two-domain behaviour.

Finally, a completely different modelling paradigm could be explored by shifting from discrete Agent-Based Models to continuous hydrodynamic approaches, such as the differential equations proposed by Toner and Tu [13] or Cavagna et al. [5]. This transition opens the door to exploiting the available empirical datasets through modern ML techniques. By applying modern procedures to discrete empirical trajectories, continuous density and velocity fields can be

extracted. Subsequently, advanced algorithms could be employed. These ML algorithms would ingest the empirical fluid-like fields to extrapolate and calibrate the exact phenomenological parameters required by the hydrodynamic models. This approach could ultimately bridge the gap between theoretical physics and real-world biological data, providing a model capable of predicting flock dynamics at a macroscopic scale.

Appendix

A Origin of the name "Boids"

The reader might be curious about the origin of the name "boids". Our own curiosity was placated by none other than Craig Reynolds himself, who so gently answered to our email regarding the matter as follows:

Subject: Origin of the name "boids"

From: Craig Reynolds

Hi Francesco and thanks for your interest in my work. I think of the term has[a] having three origins.

As you say, originally I used "bird-oid" to mean an abstract computational agent that might represent a bird in a flock, or a fish in a school/shoal, or a person in a crowd, etc. This was contracted to "boid." I wanted to express that this simple model was not meant to be a serious explanation of the complex behavior of real birds.

I liked "boids" for two reasons. It is close to how people in Brooklyn (New York) pronounce "birds" — as in that notable scene in the original movie version of The Producers.

Separate from that, Tom Duff was a famous computer graphics expert who spent time at the pioneering Graphics Lab at New York Institute of Technology (in the 1970s and 1980s) and later, many years at Pixar. When at NYIT he wrote a modeling a[nd] rendering system based on ellipsoids. He called it "soids" which I'd always thought was a fun sounding name.

Good luck with your project, and the rest of your Masters studies,
Craig

[a] Editor's note: Reynolds probably meant "as" instead of "has". *Quandoque bonus dormitat Homerus.*

B Github repository

All of the code aforementioned and more, and the data used for the analysis, as well as a complete set of graphic renditions of the flock's behaviour, are available at this github repository.

C Statement regarding the use of generative AI

In compliance with UniBo's policy regarding the use of generative AI, we declare that in the process of realizing this work, we employed Gemini 3.1 in order to optimize the code (namely by utilizing vectorization logic and speeding up the process of generating an animation) and for some other tasks of minor importance, such as helping in the betterment and debugging of parts of the code and with the grammatical revision of this document, as well as a tool for bibliographical research.

References

[1] P. W. Anderson. More is different - broken symmetry and the nature of the hierarchical structure of science. *Science*, 1972. https://inters.org/files/more-is-different.pdf.

[2] A. Cavagna et al. *The STARFLAG handbook on collective animal behaviour: 1. Empirical methods.* Animal Behaviour, 2008. https://arxiv.org/abs/0802.1668.

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